

Session 4: Operators and Expressions

Topics: Arithmetic, Logical, Relational, Increment/Decrement. Operator precedence. Example: Simple calculator for addition, subtraction, etc.

- 1) Create a simple JavaScript function called `calculateTotal` that takes two numbers: `itemPrice` and `quantity`, and returns the total bill amount using arithmetic operators.

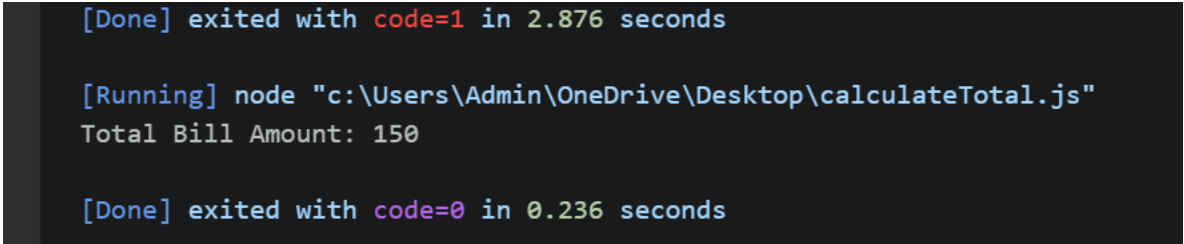
Code-

```
function calculateTotal(itemPrice, quantity) {  
    return itemPrice * quantity;  
}
```

// Example

```
let totalBill = calculateTotal(50, 3);  
console.log("Total Bill Amount:", totalBill);
```

output-



```
[Done] exited with code=1 in 2.876 seconds  
  
[Running] node "c:\Users\Admin\OneDrive\Desktop\calculateTotal.js"  
Total Bill Amount: 150  
  
[Done] exited with code=0 in 0.236 seconds
```

- 2) Build a Flipkart-style discount calculator: given product price, discount percentage, and a boolean `isMember`, use arithmetic and logical operators to calculate the final price (apply an extra 5% off if `isMember` is true).

Code-

```
#include <stdio.h>
```

```
int main() {
```

```
    float productPrice, discountPercentage, discountedPrice, finalPrice;
```

```
    int isMember;
```

```
    printf("Enter product price: ");
```

```
    scanf("%f", &productPrice);
```

```
    printf("Enter discount percentage: ");
```

```
    scanf("%f", &discountPercentage);
```

```
    printf("Is member? (1 = Yes, 0 = No): ");
```

```
    scanf("%d", &isMember);
```

```
    // Apply product discount
```

```
    discountedPrice = productPrice - (productPrice * discountPercentage /  
100);
```

```
    // Apply extra 5%% discount for members
```

```
    if (isMember == 1) {
```

```
        discountedPrice = discountedPrice - (discountedPrice * 5 / 100);
```

```
    }
```

```
    finalPrice = discountedPrice;
```

```
    printf("Final Price = %.2f\n", finalPrice);
```

```
    return 0;
```

```
}
```

Output-

```
+ FullyQualifiedErrorId : CommandNotFoundException

PS C:\Users\Admin\OneDrive\Desktop\practical> gcc discount.c -o discount
PS C:\Users\Admin\OneDrive\Desktop\practical> .\discount.exe
Enter product price: 20
Enter discount percentage: 5
Is member? (1 = Yes, 0 = No): 1
Final Price = 18.05
PS C:\Users\Admin\OneDrive\Desktop\practical> 
```

- 3) Write a function `isEligibleForOffer` that takes a user's age and total order value, and returns true if the user is 18 or older **AND** the order value is above 500, otherwise false.

Hint: Use relational and logical operators together.

Code-

```
#include <stdio.h>
#include <stdbool.h>

bool isEligibleForOffer(int age, float orderValue) {
    return (age >= 18 && orderValue > 500);
}

int main() {
    int age = 20;
    float orderValue = 750;

    if (isEligibleForOffer(age, orderValue)) {
        printf("true\n");
    } else {
```

```
        printf("false\n");
    }

    return 0;
}
```

OUTPUT-

True

- 4) Given three variables: likes, comments, and shares (all numbers), write code to check if a post is 'trending' on Instagram (at least 1000 likes OR more than 200 comments AND at least 50 shares). Print the result.**

Code-

```
#include <stdio.h>
```

```
int main() {
    int likes = 800;
    int comments = 250;
    int shares = 60;

    if (likes >= 1000 || (comments > 200 && shares >= 50)) {
        printf("Trending Post\n");
    } else {
```

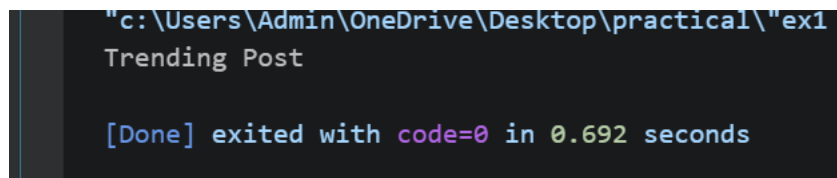
```

    printf("Not Trending\n");
}

return 0;
}

```

OUTPUT-



```

c:\Users\Admin\OneDrive\Desktop\practical\ex1
Trending Post

[Done] exited with code=0 in 0.692 seconds

```

5) Write a code snippet that demonstrates the difference between pre-increment (++count) and post-increment (count++) by logging the values before and after using both on a followerCount variable.

Code-

```

#include <stdio.h>

int main() {
    int followerCount = 100;

    printf("Initial follower count = %d\n", followerCount);

    // Pre-increment
    printf("Using pre-increment (++followerCount): %d\n", ++followerCount);
    printf("Follower count after pre-increment = %d\n", followerCount);
}

```

```
// Reset value
followerCount = 100;

// Post-increment
printf("Using post-increment (followerCount++): %d\n",
followerCount++);
printf("Follower count after post-increment = %d\n", followerCount);

return 0;
}
```

OUTPUT-

```
[Running] cd "c:\Users\Admin\OneDrive\Desktop\practical\" && gcc ex2.c -o ex2 &&
"c:\Users\Admin\OneDrive\Desktop\practical\"ex2
Initial follower count = 100
Using pre-increment (++followerCount): 101
Follower count after pre-increment = 101
Using post-increment (followerCount++): 100
Follower count after post-increment = 101
```